STUDENT ACTIVITY

TensorFlow & Data Preprocessing — Activity

Activity: Data Issue Diagnosis and Workflow Mapping

Q: Read the three data scenarios below. Identify the specific step in the Data Preprocessing Workflow (e.g., Data Cleaning, Data Transformation, Data Splitting) that addresses the issue, and provide one specific technique from the document used to fix it.

Next, in two to three sentences, explain what TensorFlow is and list three of its key features.

The Scenarios:

1. A dataset contains several records where the "Age" column is completely blank, and these records are disrupting the dataset.
2. A dataset contains raw numerical data with vastly different ranges, requiring the values to be converted into a suitable format to improve model accuracy.
3. A fully cleaned and structured dataset needs to be prepared so that a portion of it can be used to evaluate the model after it finishes learning.

Data Issue Diagnosis and Workflow Mapping

Scenario	Data Preprocessing Workflow Step	Technique Used to Fix It
1. A dataset contains several records where the "Age" column is completely blank, and these records are disrupting the dataset.	**Data Cleaning**	Handle missing values by replacing them with the **mean** or **median** (or remove records if necessary).
2. A dataset contains raw numerical data with vastly different ranges, requiring the values to be converted into a suitable format to improve model accuracy.	**Data Transformation**	Apply **Feature Scaling** (such as **Normalization** or **Standardization**) to bring all numerical values to a similar range.
3. A fully cleaned and structured dataset needs to be prepared so that a portion of it can be used to evaluate the model after it finishes learning.	**Data Splitting**	Split the dataset into **Training Set** and **Testing Set** (e.g., 80% training and 20% testing).

TensorFlow

TensorFlow is an open-source Machine Learning and Deep Learning framework developed by **Google**. It is used to build, train, and deploy artificial intelligence models for applications such as image recognition, natural language processing, and predictive analytics. TensorFlow supports both research and production environments, making it one of the most widely used AI frameworks.

Three Key Features of TensorFlow

1. **Open-source framework** for Machine Learning and Deep Learning.
2. **Supports GPU and TPU acceleration** for faster model training.
3. **Provides high-level APIs (Keras)** for building, training, and deploying neural network models easily.

Data Issue Diagnosis and Workflow Mapping

1. **Blank values in the "Age" column**
   - **Workflow Step:** Data Cleaning
   - **Technique:** Replace missing values using the **mean** or **median**.
2. **Numerical values have different ranges**
   - **Workflow Step:** Data Transformation
   - **Technique:** Apply **Feature Scaling** using **Normalization** or **Standardization**.
3. **Need a dataset for model evaluation**
   - **Workflow Step:** Data Splitting
   - **Technique:** Split the dataset into **Training** and **Testing** sets (e.g., 80:20 ratio).

TensorFlow

TensorFlow is an open-source Machine Learning and Deep Learning framework developed by Google. It is used to build, train, and deploy AI models efficiently for a wide range of applications. Its key features include **GPU/TPU support**, **Keras integration**, and **cross-platform deployment**.